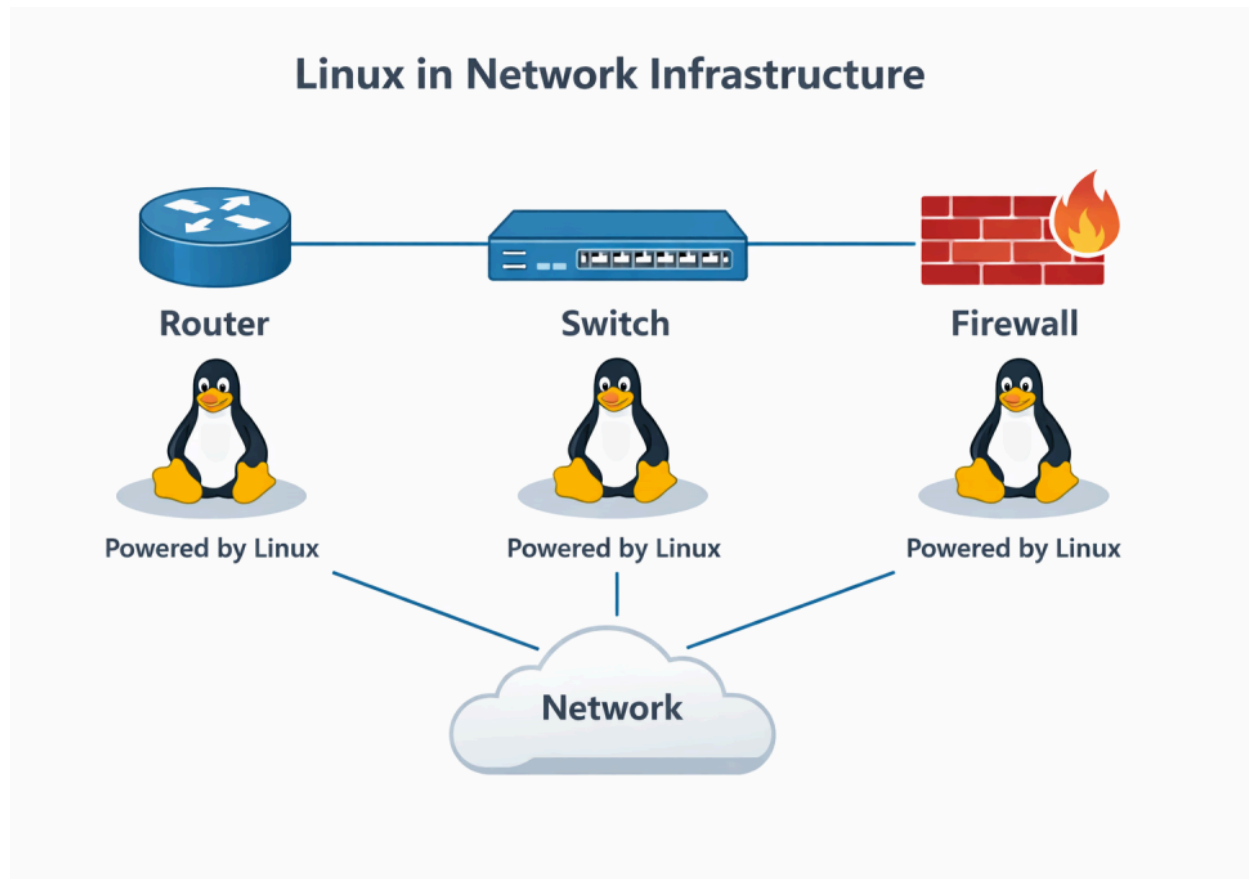


Lab 002 - Linux File System



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*Completed 07/01/26
Comprehensive Lab Report*

INTRODUCTION

The goal of this lab is to learn how to navigate a Linux system entirely from the command line.

Hardware & Materials

- Raspberry Pi Zero W
- 128 GB MicroSD Card
- Windows 11 PC/Powershell
- Home Wi-Fi

Objectives

- Finding out what my current directory is
- How Linux organizes files
- How to move around the filesystem
- How to create, rename, copy, and delete files/directories
- How to read documentation using the terminal
- Finding which OS version of Linux im running

Commands Used/Significance

Whoami - What user you are logged in as.

Pwd - What directory you are currently in.

Ls - Lists only the names of visible files and folders in rows or columns.

Ls -l - It displays files as a single vertical list containing permissions, owner, group, file size, and the last modification date.

Ls -la - gives the ordered list of file names in a directory file. Lists more information than ls and ls -l. Includes hidden files as well.

Cd - change directory

Mkdir - creates directory

Touch - creates files

Nano - write to file via GNU

Ctrl + O - save (within GNU)

Ctrl + X - exit (within GNU)

Cp - copy files

Mv - rename files

Rm - delete files (can delete directories as well using -r)

Cat /etc/os-release - list which linux version im currently using

Conclusions

After doing this lab I've successfully demonstrated:

- Linux command-line navigation
- Remote system administration via SSH
- File and directory management
- Basic Linux system exploration

Screenshots

```
student@pi-lab: ~  
  
The programs included with the Debian GNU/Linux system are free software;  
the exact distribution terms for each program are described in the  
individual files in /usr/share/doc/*/copyright.  
  
Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent  
permitted by applicable law.  
Last login: Tue Jun 30 15:12:07 2026 from 192.168.1.211  
student@pi-lab:~$ whoami  
student  
student@pi-lab:~$ pwd  
/home/student  
student@pi-lab:~$ ls  
student@pi-lab:~$ ls -l  
total 0  
student@pi-lab:~$ ls -la  
total 32  
drwx----- 4 student student 4096 Jun 30 16:15 .  
drwxr-xr-x 3 root root 4096 Jun 17 20:18 ..  
-rw----- 1 student student 143 Jun 30 16:15 .bash_history  
-rw-r--r-- 1 student student 220 Jun 17 20:07 .bash_logout  
-rw-r--r-- 1 student student 3523 Jun 17 20:07 .bashrc  
drwxr-xr-x 3 student student 4096 Jun 17 20:20 .config  
-rw-r--r-- 1 student student 807 Jun 17 20:07 .profile  
drwx----- 2 student student 4096 Jun 17 20:18 .ssh  
-rw-r--r-- 1 student student 0 Jun 30 16:15 .sudo_as_admin_successful  
student@pi-lab:~$
```

```
student@pi-lab: /  
  
student@pi-lab:~$ cd /  
student@pi-lab:/ $ pwd  
/  
student@pi-lab:/ $ ls  
bin boot dev etc home lib lost+found media mnt opt proc root run sbin srv sys tmp usr var  
student@pi-lab:/ $
```

```
student@pi-lab: ~/networking_lab
student@pi-lab:~$ pwd
/home/student
student@pi-lab:~$ mkdir networking_lab
student@pi-lab:~$ ls
networking_lab
student@pi-lab:~$ cd networking_lab
student@pi-lab:~/networking_lab$ pwd
/home/student/networking_lab
student@pi-lab:~/networking_lab$ touch notes.txt
student@pi-lab:~/networking_lab$ touch commands.txt
student@pi-lab:~/networking_lab$ ls -l
total 0
-rw-rw-r-- 1 student student 0 Jul  1 12:55 commands.txt
-rw-rw-r-- 1 student student 0 Jul  1 12:54 notes.txt
student@pi-lab:~/networking_lab$
```

```
student@pi-lab: ~/networking_lab
/home/student/networking_lab
student@pi-lab:~/networking_lab$ touch notes.txt
student@pi-lab:~/networking_lab$ touch commands.txt
student@pi-lab:~/networking_lab$ ls -l
total 0
-rw-rw-r-- 1 student student 0 Jul  1 12:55 commands.txt
-rw-rw-r-- 1 student student 0 Jul  1 12:54 notes.txt
student@pi-lab:~/networking_lab$ nano notes.txt
student@pi-lab:~/networking_lab$ cat notes.txt
Lab 002
Learning Linux commands.
student@pi-lab:~/networking_lab$ .
```

```
student@pi-lab: ~  
Lab 002  
Learning Linux commands.  
student@pi-lab:~/networking_lab $ cp notes.txt notes_backup.txt  
student@pi-lab:~/networking_lab $ ls  
commands.txt  notes_backup.txt  notes.txt  
student@pi-lab:~/networking_lab $ mv commands.txt linux_commands.txt  
student@pi-lab:~/networking_lab $ ls  
linux_commands.txt  notes_backup.txt  notes.txt  
student@pi-lab:~/networking_lab $ rm notes_backup.txt  
student@pi-lab:~/networking_lab $ ls  
linux_commands.txt  notes.txt  
student@pi-lab:~/networking_lab $ cd  
student@pi-lab:~ $ rm -r networking_lab  
student@pi-lab:~ $ ls  
student@pi-lab:~ $ cat /etc/os-release  
PRETTY_NAME="Raspbian GNU/Linux 13 (trixie)"  
NAME="Raspbian GNU/Linux"  
VERSION_ID="13"  
VERSION="13 (trixie)"  
VERSION_CODENAME=trixie  
DEBIAN_VERSION_FULL=13.4  
ID=raspbian  
ID_LIKE=debian  
HOME_URL="http://www.raspbian.org/"  
SUPPORT_URL="http://www.raspbian.org/RaspbianForums"  
BUG_REPORT_URL="http://www.raspbian.org/RaspbianBugs"  
student@pi-lab:~ $
```